

■ AgriPredict — System Architecture Blueprint

A4 Architectural Flow & Data Pipelines

■ 1. FARMER INTERFACE LAYER

■ AgriPredict PWA Web App (Next.js 16 + React 19 + TypeScript)

■ Multilingual Voice Engine (Web Speech STT/TTS in 6 Languages)

■ Auto GPS Location Detection (780+ Indian Districts Mapped)

■ State Sync & User Data ■

■ 2. APP STATE & CONTEXT CORE

■ AppContext Provider (Global State & 2-Way Location Sync)

■ LocalStorage Engine (Persistent Location, Notifications & Farm Portfolio)

■ Serverless Gateway Requests ■

■ 3. NEXT.JS 16 SERVERLESS API GATEWAY

■ /api/mandi-prices — Real-Time Mandi Rates, Volume & Best Choice Math

■ /api/chatbot — Multilingual AI Voice & Text Query Handler

■ /api/ocr — Document Scanner & Legal Text Simplifier

■ Cloud AI & Open Data Pipeline ■

■ 4. AI CLOUD & OPEN DATA SERVICES

■ Google Gemini 1.5 Flash API (Market AI, 30-Day Forecasts, Multilingual NLP, Vision OCR)

■ Agmarknet Open Data API (Govt of India Official APMC Mandi Feeds)

■ Open-Meteo Weather API (Live Temperature, Humidity, Wind & Soil Moisture Radar)

■ OpenStreetMap Nominatim & BigDataCloud (Browser GPS Geocoding Engine)

■ A4 Poster Specifications: 210mm x 297mm Standard A4 Ratio | Built for AgriPredict Platform Architecture